

Project Proposal

Project Proposal: Modeling Strategic Decision-Making in Poker using a Nano-GPT Architecture

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1. Problem Description

The objective of this project is to build a decoder-only Transformer model from scratch that can predict the next action in a game of Texas Hold'em Poker. Instead of predicting the next letter in a word, the model will predict the next **strategic move** (e.g., Check, Bet, Raise, or Fold) based on the "Hand History"—the sequence of previous cards dealt and bets made.

This project explores whether a generative AI architecture, typically used for natural language, can internalize the rules of a game and mimic the betting patterns of human players.

2. Dataset

I will use the **Poker Hand Histories** dataset (sourced from Zenodo or Kaggle).

- **Source:** [Zenodo Poker Hand Histories](#).
- **Description:** The data consists of text-based logs of millions of poker hands. These logs include the players' hole cards, the community cards (Flop, Turn, River), and every betting action taken during the hand.
- **Preprocessing & Tokenization:** I will implement a custom tokenizer. Instead of character-level encoding, I will use a **word-level tokenizer** where each unique action (e.g., `Bet`, `Fold`, `Raise`) and card (e.g., `Ah` for Ace of Hearts) is treated as a single token. This transforms a poker game into a "sentence" of strategic tokens.

3. Modeling Algorithms (The "Nano-GPT" Architecture)

The model will be built using a **Decoder-only Transformer** architecture implemented in PyTorch/NumPy. Key components built from scratch include:

- **Causal Self-Attention:** This allows the model to "attend" to critical previous events, such as a large raise in a previous round, to decide the current action.
- **Positional Encodings:** To help the model understand the sequence of rounds (Pre-flop vs. River).
- **Multi-Head Attention Layers:** To enable the model to simultaneously track different aspects of the game, such as card strength and opponent betting size.
- **Linear & Softmax Output:** To generate a probability distribution over the possible next actions.

4. Expected Outcomes

- **Action Accuracy:** I will measure how often the model correctly predicts the action taken by a human player in the test set.
- **Strategic Coherence:** I will evaluate if the model learns "logical" betting. For example, a successful model should never predict a "Check" after an opponent has "Bet" (as that is an illegal move).
- **Error Analysis (Illegal Moves):** Similar to a confusion matrix, I will track "Illegal Move Rates." If the model predicts an action that is impossible given the current game state, it indicates a failure in learning the game's underlying "grammar."
- **Generative Play:** I will provide the model with a "starting hand" and simulate a full game to see if the generated betting sequence resembles a realistic poker strategy.